# Rishi Dave

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## Education

**University of California, Riverside** — Riverside, CA

*B.S. + M.S. Computer Science, Minor in Mathematics: GPA: 3.97/4.00 | Chancellor's Honor List* — *Expected June 2028*

- Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Operating Systems, Distributed Systems, Computer Networks

## Experience

**Capital One: Decision the Deal (Financial Services)** — June 2026 – Present

*Software Engineering Intern* — *Dallas, TX*

- Extending a **graph-based loan-decisioning engine** in **Java/Spring Boot** that compiles business-analyst policy rules into executable graph nodes spanning REST service calls, ML inference, and external integrations across **AWS Neptune, DynamoDB, S3, and PostgreSQL**.
- Architecting **multi-graph tenancy** to let one engine serve multiple policy graphs by extending the database layer, building graph-routing logic, and validating with a shadow-graph migration of the legacy hard-coded policy engine.

**Microsoft ONNX Runtime (17K+ stars): Open-Source Contributor** — Jan. 2026 – Present

*ML Graph, Quantization & CUDA Optimizer | **45 PRs** (38 merged)* — *Remote*

- Reduced inference graph complexity by **37%** on Qwen3-0.6B (~1,500 → ~950 nodes) with **operator-fusion and constant-folding** compiler passes, and restored **GPT-2 attention fusion** (broken since 2023) using multi-variant subgraph pattern matching with golden-file regression tests.
- Closed multiple correctness gaps in the **INT4/INT8 quantization (QDQ)** pipeline — MatMulNBits fusion, N-D weight support, and calibration caching — affecting every user running `quantize_static` on modern quantized models.
- Merged a **C++ core-graph** fix (after a 70-day review) propagating outer-scope type info to subgraph nodes, unblocking BeamSearch/Whisper ops, and added **opset-21/23 CUDA kernel registrations** to keep models on-GPU. Reviewed by Microsoft's transformer-optimization lead.

**Meta PyTorch (85K+ stars): Open-Source Contributor** — Feb. 2026 – Present

*Distributed Training & Optimizer Infrastructure | **6 PRs** (3 merged, 3 in review)* — *Remote*

- Fixed **distributed gradient-accumulation crashes** in FSDP2 and mixed-precision training by registering a missing sharding strategy in PyTorch's **DTensor** subsystem; modernized 112 lines of legacy **ZeRO-optimizer** broadcast with PyTorch's built-in distributed API.

**UC Riverside: Information Technology Solutions** — June 2025 – Aug. 2025

*Data Science Fellow ($5,500 stipend)* — *Riverside, CA*

- Cut academic-planning time from **45 min to <2 min** for **500+ students** by building a full-stack tool (**React**, **FastAPI**, **MongoDB**, **PostgreSQL**) with a LangGraph AI planning agent. Shipped to production via Docker and CI/CD.
- Built a **CatBoost ranking model** predicting course enrollment within **15% MAPE** across 10,000+ sections, plus **ETL pipelines** and a **RAG retrieval** layer over a **Pinecone** vector database for semantic search at <100ms latency.

## Research

**Cooperative 3D Perception: Solo Researcher** | Prof. Hang Qiu, UCR — Sep. 2025 – Present

- Improved 3D detection accuracy by **56%** (AP@0.7: 0.43 → 0.67) across **70+ ablation experiments** (300+ GPU-hours, NVIDIA A6000) by designing multi-scale BEV feature fusion, analytical SE(2) pose alignment, and gated residual connections. Wrote a **dependency-free PyTorch** reimplementation of Point Transformer V3, transferring pretrained weights without C++ extensions.

**FPGA-Accelerated ML Inference: First Author** | Prof. Phillip Brisk, UCR — Feb. 2025 – Present

- Achieved **491× CPU throughput** (0.104ms latency) and **1,069× energy efficiency** with zero accuracy loss by engineering a fused single-pass **C++** kernel (II=1) and a dot-product reuse engine, scaled to 3 compute units at 397 MHz. **Accepted as poster at IEEE FCCM 2026**.

## Projects

**MUTA: Scalable ML-Serving Backend** | *FastAPI, PostgreSQL, Redis, LightGBM, AWS ECS, Docker* — Nov. 2025

- Served **20+ weekly users** at **<200ms** latency by building an ML scheduling service on **AWS ECS**: a LightGBM model with automated retraining and drift detection, **Redis** caching, Celery task queues, **Prometheus** monitoring, and zero-downtime CI/CD.

## Technical Skills

**Languages**: Python (advanced), C/C++, Java, Go (familiar), SQL, JavaScript/TypeScript, Bash

**Systems & ML**: PyTorch, ONNX Runtime, CUDA, distributed training (FSDP, DTensor, ZeRO), compiler/graph optimization, NumPy, TensorFlow

**Web & Cloud**: React, Next.js, FastAPI, Spring Boot, Node.js, PostgreSQL, MongoDB, Redis, AWS (Neptune, DynamoDB, S3, ECS), Docker

**Practices**: REST APIs, distributed systems, CI/CD (GitHub Actions), Git, Linux, unit/regression testing, code review